# SkillSight

## Detecting Explicit vs. Implied Skills in Text (0.5 / 1)

Presented by: Yonatan Elman, Michael Kovalchuk, Roni Fadlon

# Project Review

**Motivation**

Extract CS/Hi-Tech skills from resumes with Explicit/Implicit labels (from a predefined Global Skill Vector).

1. Vector → Sparse list: only relevant skills are stored, labeled 1.0 / 0.5.
2. We added skill aliases (for explicit mentions).
3. We defined a fixed global vector of 110 CS-related skills.
4. We switched from 0/1/2 to 1.0/0.5, where 0 means the skill is not present / not predicted.

**Novelty**

The model extracts implicit skills from context, not only explicit keywords.

# Dataset

- Synthetic JSONL dataset: job_description (4–6 sentences) + skills (3–6 skills) from a predefined GLOBAL_SKILL_VECTOR.
- Labels per skill: 1.0 = Explicit, 0.5 = Implicit.

- Generated with LLM (gpt-4o mini) using a structured prompt
- aliases for Explicit skill.
- Post-generation: automatic validation & repair to prevent leaks/missing explicit mentions.
- 473 synthetic samples
- 110 skills in the Global Skill Vector

# EDA

## EDA Summary

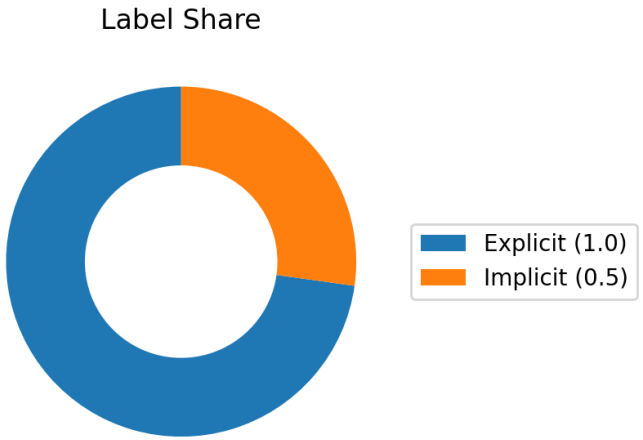
Samples: 473

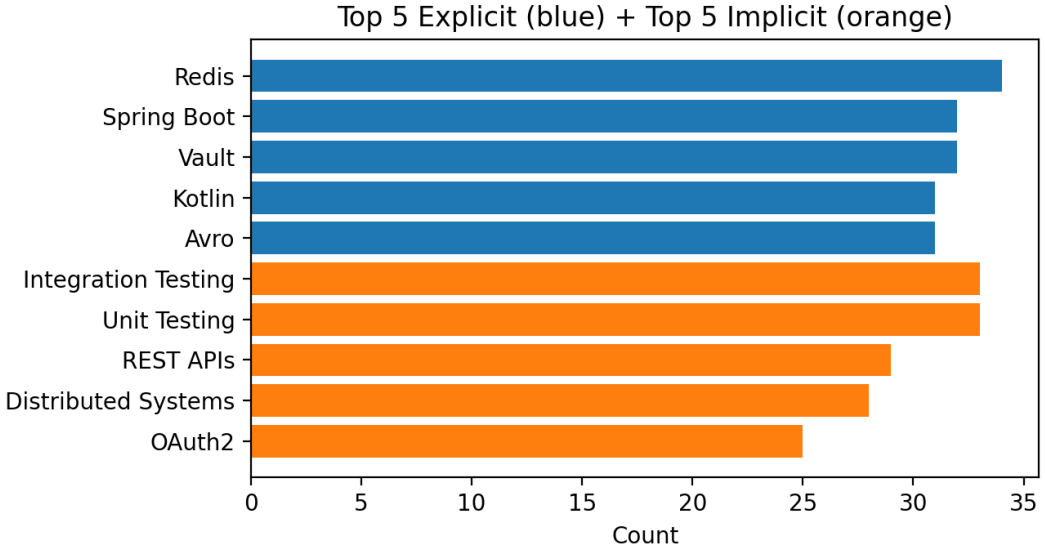
Global skills: 110

Total labeled skill occurrences: 2318

Explicit vs Implicit: 72.8% / 27.2%

Samples with 0 implicit: 20.9%




**Quality (Types) + Length**

Missing keys: 0 (0.0%)
Bad skills type: 0 (0.0%)
Duplicates: 0 (0.0%)
Non-ASCII: 1 (0.2%)
Double spaces: 0 (0.0%)
Control chars: 0 (0.0%)
Unknown skills: 1
Bad label values: 0

Sentence count: mean 5.53, median 6.0, min/max 4/8
Words: mean 92.4, median 91.0, min/max 60/131
Chars: mean 701.6, median 696.0, min/max 474/1009

Imbalance: zero-occ 1, <10 4, <20 44, Top10 cover 14.15%

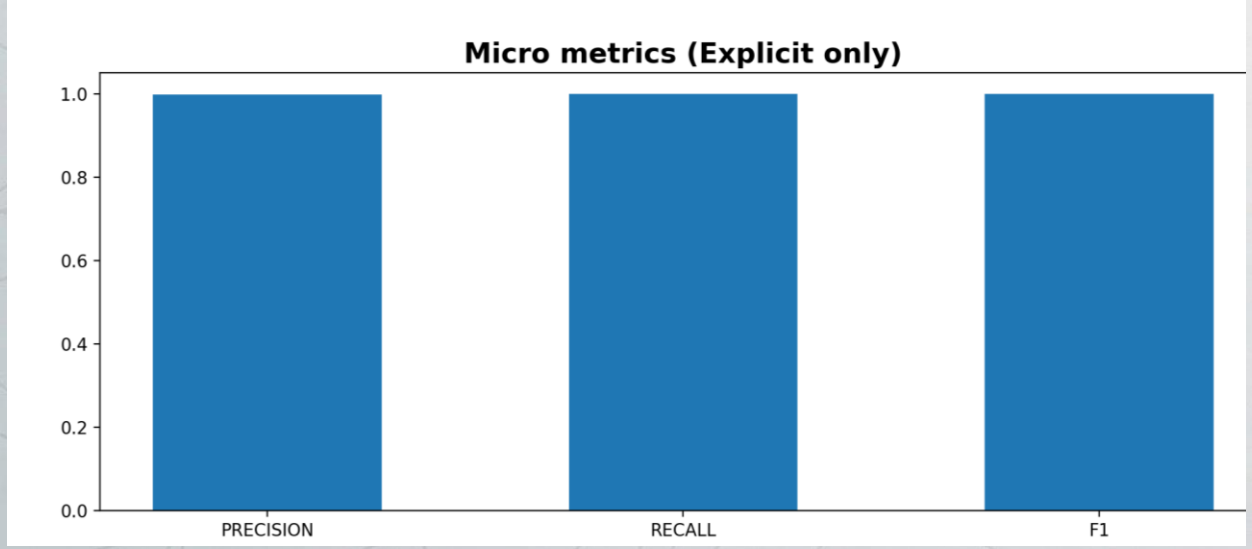
Co-occurrence: Top pair (Integration Testing, Unit Testing) appears 11/473.

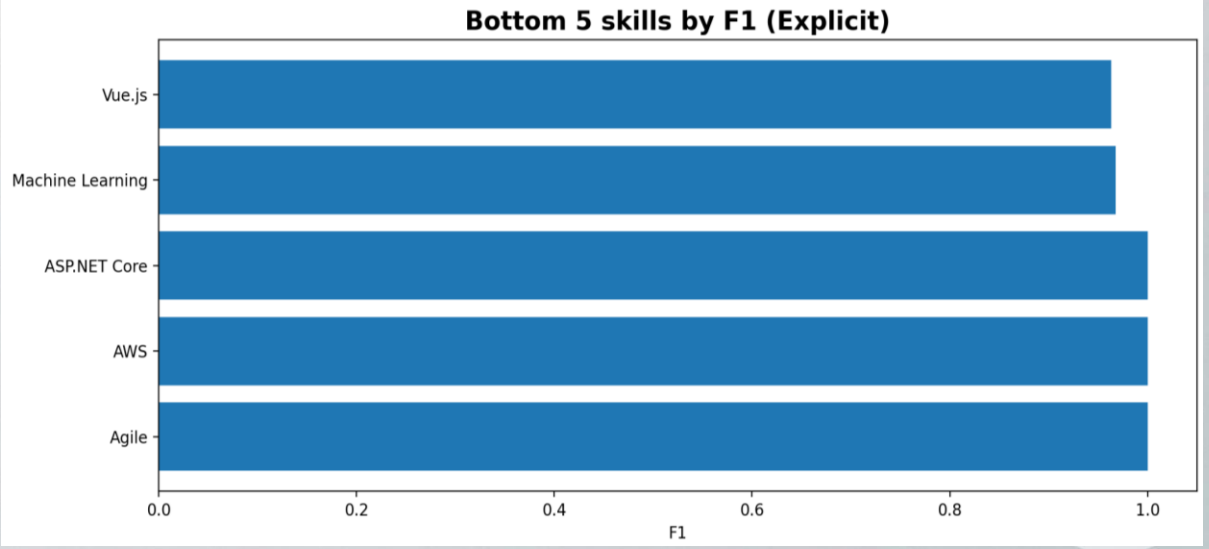
# Baseline 1- keyword matching

- Type: keyword matching (aliases + word-boundary regex)
- Output: Explicit only (1.0 if match else 0.0)
- N=473 | Micro F1=0.999 | TP/FP/FN=1688/2/0
- Errors: FP=2, FN=0
- High explicit scores are expected because labels rely on alias presence.

FP #1 — Machine Learning (gt=0.5 → pred=1.0)
Trigger: “machine learning initiatives”
Reason: Explicit-only baseline

FP #2 — Vue.js (gt=0.5 → pred=1.0)
Trigger: “Vue.a”
Reason: spurious match



# Baseline 2- zero shot

- Type: Zero-shot classification (LLM + prompt) over GLOBAL_SKILL_VECTOR
- Output: predicts 0.5 (Implicit) / 1.0 (Explicit)
- N=473

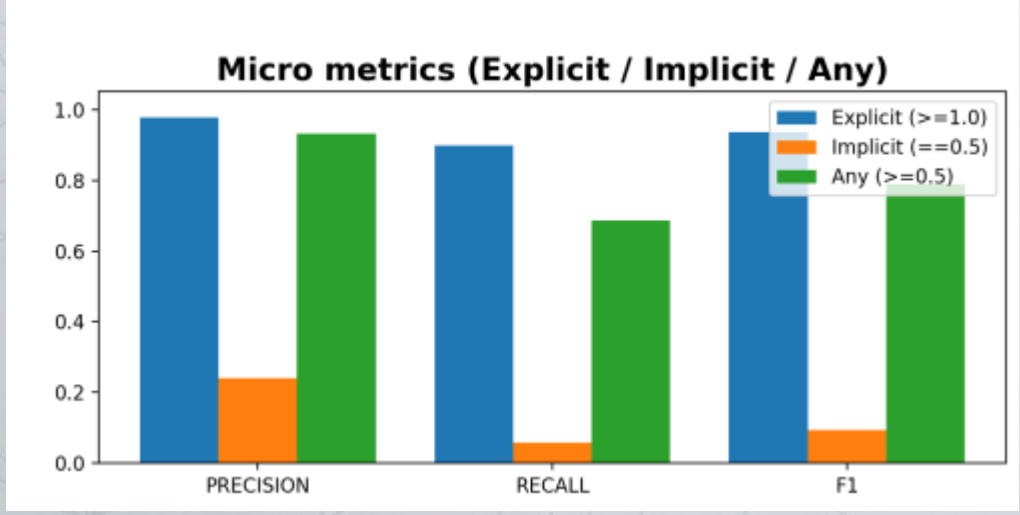
Micro F1:

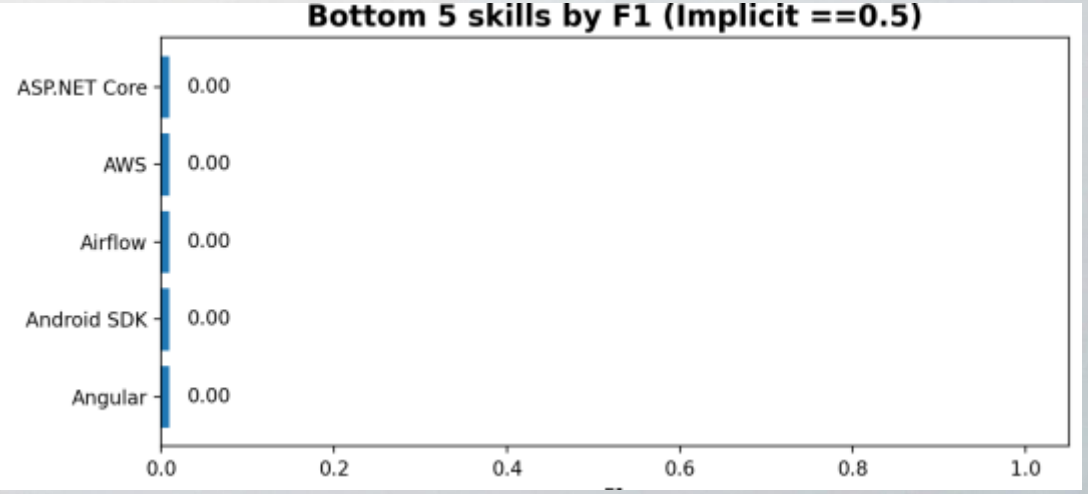
- Implicit (==0.5): 0.094 (P=0.240, R=0.059)

Key takeaway: Strong on Explicit, very weak on Implicit (low recall).

FN #1 — Load Balancing (gt=0.5 → pred=0)
Trigger: “peak traffic”, “rate limiting”
Reason: Implicit-only cue (no explicit mention)

FP #1 — Tailwind CSS (gt=1.0 → pred=0.5)
Trigger: “using Tailwind …”
Reason: Explicit mention, but model downgraded to implicit (uncertainty/noisy phrasing)



# Plan

| Step | When | What we will do (Scope) | Expected outcome |
|---|---|---|---|
| 1 | Week 10 | Generate 1,500–2,000 synthetic samples; strengthen implicit cues; reduce label leakage; rebalance 0/0.5/1 | Larger dataset (1,500–2,000) + cleaner implicit ground truth |
| 2 | Week 10 | Re-run EDA + baselines on updated data | Updated metrics + identified failure modes |
| 3 | Week 10 | Stronger baseline inference (zero-shot with stronger LLM) | Better reference baseline for implicit |
| 4 | Week 11 | Build and fine-tune a classifier for 0/0.5/1 using BERT/RoBERTa | Build a dedicated model for explicit vs implicit labeling |
| 5 | Week 11 | Error analysis iteration (top errors + skill-specific fixes) | Targeted improvements + final evaluation |
| 6 | Before deadline | Prepare final presentation + report | Final submission package |